Sovereign AI Federation v2

SIGMA AI-assisted research synthesis

Needs review: TeX verification is incomplete. Review the archived proofreading diagnostics before relying on the typesetting.

AI-generated research synthesis · SIGMA

Abstract

The investigation asks: Can universities with uneven GPU capacity provide useful, reliable AI research access while retaining control of sensitive data and staying within hard budgets? The frozen record includes a source-linked finding that is not independently verified here: With n_G,local,S=200, the expected sensitive gate μ_G,S≥160 requires q_G,local,S≥0.80. The analysis distinguishes recorded arguments from demonstrated results, identifies remaining gaps, and does not claim experiments or proof steps absent from the frozen record.

About this paper

This paper was generated by SIGMA, an AI-assisted research system, from the project's recorded investigations, findings, and supporting evidence. The analysis and conclusions reflect an automated synthesis of that research. Although SIGMA incorporates evidence tracking, critical review, and consistency checks, the resulting paper has not necessarily undergone independent verification or peer review. Readers should evaluate its arguments, supporting evidence, and stated limitations before relying on its conclusions. Research style: Define the workload, outcome oracle, unit of analysis, execution horizon, and evaluators before drawing reliability conclusions. Distinguish formal conditional guarantees, counterexamples, saved incidents, observed benchmarks and proposed fault-injection studies. Document retracted generalizations, correlated repetitions, sampling assumptions and what external evidence would validate a claimed improvement.

Evidence coverage limitation

The evidence index omitted 9706 material saved research field fragment(s) across 10 source(s). These fragments were not reviewed for this paper. Interpretations that could depend on omitted material remain provisional; the complete frozen source snapshot and omission ledger are retained with the generation record. *Claims discussed:* C6

1 Introduction and problem statement

The investigation asks: Can universities with uneven GPU capacity provide useful, reliable AI research access while retaining control of sensitive data and staying within hard budgets?

Claims discussed: C1

2 Methods and evidence

This synthesis organizes the project's frozen research records and their saved evidence references. It does not claim new experiments.

3 Background and research context

The central research question is whether universities with uneven GPU capacity can provide useful AI research access while preserving campus authority over data and expenditure; federation is one candidate architecture, not the assumed answer. The relevant comparison therefore concerns authorized, useful research outcomes rather than aggregate GPU capacity or nominal inference throughput. The recorded propositions establish three distinct questions: whether a request may be authorized, whether its financial exposure can remain within hard caps, and whether additional execution destinations improve service enough to justify their costs and obligations.

These questions must remain separate because neither authorization safety nor budget preservation establishes task quality, availability, or a federation advantage [C1–C5; [Research session 1]].

MindRouter supplies the intended campus-gateway starting point. The supplied research attributes protocol translation across OpenAI, Ollama, and Anthropic clients and Ollama and vLLM backends, weighted deficit round-robin scheduling, per-user quotas, audit logging, and GPU telemetry to its published gateway description. However, neither `https://mindrouter.ai/documentation.html` nor `https://ai4ra.uidaho.edu/wp-content/uploads/2026/04/Sheneman_MindRouter_PEARC26_Short_Paper.pdf` was retrieved in the supplied research; October 31, 2026 is a project-reported historical review date, not a newly established access date.

These capabilities consequently remain attributed descriptions rather than independently measured facts, and cross-campus federation, frontier routing, and hierarchical financial reservations are hypothetical extensions rather than verified MindRouter features [C2, C5; [Research session 1, artifact 2]; [Research session 1, artifact 3]].

The comparison distinguishes campus-local inference only, approved frontier API access only, a campus gateway with separately approved external access, and a proposed federation added to that gateway arrangement. The gateway-plus-separate-access baseline matters because it tests whether cooperation contributes something unavailable through a smaller set of institutional relationships. In a meaningful federation comparison, the gateway and federation alternatives retain identical non-federation entitlements, local allocations, and paid-access limits; additional partner capacity

and its operating and agreement costs are explicitly attributed to federation.

Separately approved external access is not automatically financially mediated and therefore cannot inherit a hard-cap guarantee from approval alone [C2, C5; [Research session 1, artifact 5]; [Research session 1, artifact 6]].

The proposed workload spans classroom coding, routine analysis, demanding scientific reasoning, multimodal or long-context work, agentic workflows, sensitive analysis, and reproducible batches. Its unit of analysis is a registered parent research task with an input snapshot, requested output, policy classification, acceptance rubric, deadline, provenance requirements, and total resource envelope; calls, tools, retries, and reproducibility repetitions are descendants rather than additional successes. The outcome oracle requires authorized processing and satisfaction of prespecified quality, deadline, and provenance requirements, with correct refusals reported separately from delivered research results.

A bounded 90-day pilot is proposed as an execution horizon, not reported as completed work [C4, C5; [Research session 1, artifact 5]; [Research session 1, artifact 6]].

Evaluation is intended to combine held-out coding tests, known-result and unit checks for analysis, blinded faculty review of scientific reasoning, labeled multimodal fixtures, and agent root-goal and tool-boundary checks. Faculty and students must approve the quality rubrics, latency targets, provenance requirements, and accessibility needs before those criteria can support a service judgment. Overall completion coverage must retain submitted demand in its denominator, while model-quality comparisons use matched policy-eligible strata; otherwise an API-only alternative could appear favorable simply by excluding restricted work.

Sensitive-task coverage must also remain separately visible, because the recorded inference from aggregate local usefulness to sensitive-stratum feasibility was withdrawn [C4, C5; [Research session 1, artifact 2]; [Research session 1, artifact 8]].

Campus sovereignty enters the proposed design as an eligibility constraint before any destination is ranked by quality, deadline feasibility, or cost. The home campus determines identity, classification, approved processing domains, retention, contracts, audit requirements, and available quota, including restrictions inherited by prompts, outputs, logs, embeddings, and caches. Ambiguous classification supplies no execution destination until resolved, and failure, overload, cancellation, or a better model score supplies no new authority to disclose content.

This is an enforcement requirement across tools, network egress, storage, and logging, not a property established by a routing recommendation [C1–C3; [Research session 1, artifact 3]].

The first conditional proposition identifies a limit on disconnected authorization. Suppose a remote admission point has an earlier campus approval, receives no subsequent authoritative information through any channel, and considers a fresh request without a previously committed request-specific grant. Its observations are identical whether the home campus leaves approval unchanged or revokes it during the partition; admitting in the former execution therefore also admits in the latter under the same decision rule.

Under this model, availability for fresh disconnected admissions conflicts with immediate revocation enforced at the authorization-decision ordering point, including for randomized rules when safety

requires zero unauthorized-admission probability [C1; [Research session 1, artifact 1]].

The strongest objection to a broad reading of that proposition is that not every federated request is freshly authorized while disconnected. A reachable home authority removes the missing-information premise, while a campus-approved lease explicitly permits weaker revocation timing; a cache lifetime alone does not establish such permission. An earlier request-specific grant also differs from fresh admission, and authorization before revocation can permit later transmission under decision-time semantics.

Thus the recorded limitation neither identifies a MindRouter defect nor establishes a guarantee against post-revocation delivery, execution, or retention [C1, C4; [Research session 1, artifact 1]; [Research session 1]].

The second proposition concerns financial exposure rather than data-export authority. For each applicable user, project, department, and campus scope, the proposed ledger maintains the invariant $L_s + H_s \leq B_s$, where L_s denotes settled charges, H_s unresolved reservations, and B_s the cap. Conditional preservation requires complete mediation of billable actions, enforceable finite liability bounds, and durable atomic reservations across all scopes; settlement replaces a hold only with an authoritative final charge no greater than that hold.

Timeout or cancellation alone leaves possible liability unresolved, so releasing its reservation can admit additional spending against money already exposed [C2; [Research session 1, artifact 3]].

The related enforcement proposition adds assumptions that cannot be inferred from root-task accounting alone: a linearizable home authority, trusted binding of immutable request content, mediated one-use issuance, and campus acceptance of decision-time authorization. The missing historical originals prevent endorsement of an exact correspondence between the successive specifications, while the supplied excerpts support treating these controls as proposed premises rather than implemented properties. In particular, replay isolation after recovery does not itself preserve financial exposure, and a current zero-charge observation does not establish finality if an earlier sender can still incur liability.

These objections narrow the conditional argument without implying that reservations, quotas, or scheduling weights establish availability or starvation freedom [C2, C3; [Research session 1, artifact 7]].

Service evidence and safety evidence consequently require different evaluators. A paired task comparison can assess useful-completion differences only within its declared population and sampling model; shared queues, outages, caches, budgets, and repeated templates can make nominal repetitions correlated rather than independent observations. Such comparisons do not establish sender safety, which instead requires inspection and fault-injection evidence at authorization, egress, recovery, and billing boundaries.

The supplied research contains conditional arguments and hypothetical counterexamples, not observed workload benchmarks or deployment incidents demonstrating federation reliability; even finite tests with no observed violations would provide bounded support rather than universal safety [C3, C4; [Research session 1]].

Finally, the recorded zero-additional-usable-peer-capacity scenario defines a boundary case, not

a general finding against federation. Equal useful service opportunities, no offsetting benefit, and strictly positive federation overhead favor the non-federated gateway only under those stipulated conditions. Establishing an improvement requires campus measurements of policy-permitted, model-compatible, reachable, deadline-feasible partner service, together with useful-task outcomes and non-overlapping cost evidence that includes administrative obligations.

The evidence therefore changes the central question from whether campuses can pool GPUs to whether cooperation adds useful, authorized service beyond campus provision and separately approved external access without weakening the authority and liability conditions that make that service acceptable [C5; [Research session 1, artifact 8]].

Claims discussed: C1, C2, C3, C4, C5

4 Recorded findings and observations

The recorded findings narrow the case for sovereign AI access by showing that aggregate completion rates, authorization decisions and budget invariants establish different—and non-interchangeable—properties. The strongest substantive correction concerns sensitive workloads: an acceptable overall local completion fraction does not establish useful service for campus-only tasks. The other findings identify conditional accounting guarantees and counterexamples to stronger authorization, recovery and scheduling claims, rather than demonstrated federation performance. [C6–C10]

The published-capability boundary is consequential to interpreting these findings. The supplied description attributes OpenAI/Ollama/Anthropic client translation, Ollama/vLLM backends, WDRR scheduling, per-user quotas, audit logging and GPU telemetry to MindRouter's campus gateway; these are descriptions of published claims, not independently measured capabilities in the supplied record. The documentation and PEARC'26 short paper were not retrieved for this investigation, so the reported starting-reference review date of 2026-10-31 is not a newly established access date, and no changed claim has been verified.

Cross-campus federation, frontier routing and hierarchical financial reservations therefore remain proposed extensions, not findings about existing MindRouter functionality. [C7, C8]

The numerical findings concern a hypothetical monthly workload of 2,000 root research tasks, including 200 sensitive campus-only tasks and 1,800 API-eligible tasks. The four alternatives are campus-local inference only, L; approved frontier API only, A; a campus gateway with separately approved external access, G; and a hypothetical federation extension, F. The worksheet assumes deadline-feasible local admission of 1,000 roots, allocated to 200 sensitive and 800 other tasks under G. These quantities specify a scenario, not observed campus demand, measured capacity or institutional policy. [C6, C8]

The unit of useful service is a completed root research objective, not an inference call, retry or agent exchange. In the repaired worksheet, usefulness is the joint event that processing is authorized and the final result meets its quality, deadline and provenance requirements; its probability is not obtained by multiplying independently assumed component probabilities. The proposed evaluation uses task-specific tests, numerical checks or expert assessment, with two blinded assessors and a

third adjudicator for disagreements, but these evaluators have not produced recorded outcomes. The separate bounded pilot has a proposed horizon of 90 days; it must not be confused with an executed validation of the monthly worksheet. [C6, C10]

For a stratum and destination, let n_{bds} denote distinct admissions under baseline b, and let q_{bds} denote their average marginal probability of joint usefulness. Expected useful completions are then $\mu_b = \sum_{s,d} n_{bds} q_{bds}$. This expectation identity does not require independent task outcomes, but neither does it establish a probability of meeting a realized service target. Calls and reproducibility repeats remain descendants of their parent task, while correct policy refusal is recorded separately rather than counted as delivered research. [C6, C10]

The sensitive-coverage correction follows directly from disaggregating the local expectation. With the stated admissions, aggregate local usefulness of 0.80 imposes only $200q_{G,local,S} + 800q_{G,local,N} = 800$, where S and N distinguish sensitive and other local tasks. It does not identify either stratum's usefulness: complete failure on sensitive work can coexist with the stipulated aggregate if every other admitted local task succeeds. Sensitive-first admission does not remove this counterexample, because priority determines access to execution rather than the usefulness of the resulting artifacts. The earlier unqualified inference from aggregate local usefulness to sensitive feasibility is therefore withdrawn. [C6]

The repaired sensitive gate requires an additional stratum-specific premise. When $n_{G,local,S} = 200$, the expected target $\mu_{G,S} \geq 160$ requires $q_{G,local,S} \geq 0.80$; with only $a_S \leq 200$ sensitive admissions, the condition becomes $a_S q_{G,local,S} \geq 160$ and is impossible when $a_S < 160$. A simple repair sets both local stratum probabilities to 0.80, preserving the worksheet's 800 expected local completions. More generally, preserving that aggregate while satisfying the sensitive threshold requires $q_{G,local,N} = (800 - 200q_{G,local,S})/800$. The accompanying table illustrates how the same aggregate can conceal failure or satisfaction of the sensitive gate. [C6]

This repair changes the logical status of feasibility, not the empirical evidence for it. The assumed sensitive probability is additional and hypothetical; it is neither implied by aggregate local performance nor established by allocating all sensitive roots locally. Consequently, an expected-count calculation cannot support a realized coverage guarantee or a campus-specific least-cost-feasible designation. Evidence capable of supporting that designation would need admissions and joint useful completions by classification and task stratum, including failures, deadline misses and unserved demand, rather than only an overall success fraction. [C6]

The accounting finding is similarly conditional but addresses a different outcome. For each applicable budget scope s, the abstract ledger maintains settled charges L_s and unresolved reservations H_s under cap B_s, with $L_s + H_s \leq B_s$. Atomic admission preserves the inequality by reserving a finite enforceable maximum charge r before dispatch; authoritative settlement replaces that reservation with final liability z, where $0 \leq z \leq r$, so covered exposure cannot increase. This reasoning requires complete mediation of billable paths, durable concurrent updates and enforceable charge bounds, including tools, retries and separately approved accounts.

It is an abstract allocation guarantee, not verification of provider billing or an implemented financial control. [C7, C10]

The timeout counterexample establishes why unresolved liability cannot be released merely because

a client stops waiting or requests cancellation. Under a cap of 10, releasing an ambiguous liability of 6 and admitting another liability of 6 permits total exposure of 12. A separate recovery counterexample shows that fresh identifiers and credentials do not repair a stale backup that has forgotten the first liability: replay isolation and preservation or conservative reconstruction of outstanding exposure are both necessary. These are hand-worked counterexamples to unsafe rules, not saved operational incidents or executed fault-injection results. [C9, C10]

Final reconciliation also has to account for delayed sending, not merely charges observed so far. Under decision-time authorization, a grant committed before revocation or cancellation may still lead to later transmission, receipt or execution; the recorded design does not establish a stronger post-revocation non-delivery guarantee. Thus a current zero-charge report, local silence or an ended invoice period cannot by itself justify releasing exposure while an authorized continuation can still incur liability. The missing operational final-liability criterion and incomplete earlier transition definitions prevent treating the proposed refinement as a verified implementation, even where its local accounting arithmetic is sound. [C9]

The scheduling findings impose another important separation: WDRR weights do not establish unconditional latency bounds or starvation freedom. The recorded counterexamples preserve the need to examine compatible capacity, model and memory fit, blocking, cold starts and actual admitted workload rather than treating scheduling weights as a service guarantee. Likewise, retaining uncertain financial holds can protect caps while reducing availability and fairness. Budget preservation and fair useful access therefore require separate evaluation; neither follows from the other. [C7, C10]

The proposed federation-benefit gate was deliberately strengthened, rather than retrospectively credited with a stronger conclusion. A point estimate of at least 10 percentage points with an interval excluding zero supports a different proposition from a confidence interval whose lower endpoint exceeds 0.10; only the latter is the proposed gate for evidence of a benefit exceeding that threshold. The exploratory target of 30 parent tasks per stratum was not justified as a precision target, and no completed power calculation establishes an affordable enrollment size for the revised gate.

Its probability interpretation depends on independent, identically sampled parent-task pairs within the declared population; correlated repetitions do not enlarge that sample, and an independent-block analysis would target a separately defined block-average effect. [C10]

The resulting change to the central argument is substantive: aggregate throughput is no longer sufficient evidence of acceptable sovereign research access. The numerical case now requires explicit sensitive-stratum usefulness, the budget case requires preserved unresolved exposure, and the authorization case stops at an accepted decision-time ordering point. A claimed federation improvement would therefore have to appear as additional policy-permitted, rubric-passing, on-time and provenance-complete root outcomes under matched entitlements, with measured costs and separate safety and service gates—not as nominal GPU capacity, repeated calls or a stronger interpretation of the earlier aggregate worksheet. [C6–C10]

Table 1: Local stratum configurations consistent with the same aggregate expectation. Each row assumes 200 sensitive and 800 other local admissions and yields 800 expected useful local completions. These are hypothetical probabilities and direct consequences of the recorded expectation equation, not measurements or realized-service guarantees.

Configuration	**Sensitive usefulness** $q_{G,local,S}$	**Other local usefulness** $q_{G,local,N}$	**Expected useful sensitive roots**	**Expected sensitive gate** $\mu_{G,S} \geq 160$
Recorded counterexample	0	1	0	Fails despite the stipulated aggregate
Explicit symmetric repair	0.80	0.80	160	Meets the expected threshold
Boundary case of the recorded general relation	1	0.75	200	Exceeds the expected threshold

Evidence: [Research session 1, artifact 8]; claims: C6

Claims discussed: C6, C7, C8, C9, C10

5 Research methods and proposed approaches

The proposed method separates two requirements that the federation argument must satisfy: useful research completion at the task level and valid authorization at the dispatch level. It compares campus-local inference, approved frontier API access, a campus gateway with separately approved external access, and a hypothetical federation added to that gateway arrangement. The recorded work comprises workload specification, conditional scenario calculations, an abstract transactional design and hand-worked adversarial schedules; it does not comprise an executed comparative benchmark or a deployed federation. Claims C12 and C11 respectively define the evaluation unit and the authorization semantics needed to make that comparison meaningful.

The common workload is a hypothetical monthly cohort of 100 distinct active researchers submitting 2,000 root tasks. The accompanying table specifies the task allocation, proposed deadlines and acceptance instruments. These are disjoint accounting bins rather than a taxonomy in which sensitivity is a separate technical capability: a coding or analysis task can also carry restrictions, but each root is counted only once in this scenario. The monthly horizon governs workload accounting; annual equivalents assume 12 identical months and do not constitute additional observations (C12).

A useful completion must jointly satisfy authorization, the end-to-end deadline, the frozen task rubric and the absence of critical factual or security errors. Held-out tests, known-result checks, blinded scientific review and labeled fixtures provide task-specific outcome oracles; agentic work additionally requires root-goal and tool-boundary checks. Batch acceptance requires complete provenance and an explicitly specified numerical tolerance, not an assumption that an API can reproduce an output bit for bit. Correct refusals are recorded separately from delivered research, and the deadlines, rubrics and evaluator arrangements remain subject to campus approval (C12).

The accounting hierarchy distinguishes a root research task, R, from a logical inference or tool operation, O, and a potentially billable invocation attempt, A. Calls, paired Investigator/Skeptic exchanges and retries remain descendants of their original root rather than becoming additional useful completions. This distinction also prevents workflow amplification from disappearing from the comparison: additional calls can increase resource use and financial exposure without increasing the number of completed research goals (C11; C12).

Policy eligibility is assessed separately from technical suitability and deadline-feasible capacity. In the illustrative approval matrix, all 2,000 roots are permitted locally, 1,800 non-sensitive roots are eligible for a named API destination, and only coding, routine analysis and reproducible batches—1,100 roots—are eligible for a named partner campus. These are scenario assumptions, not established institutional permissions, and federation does not itself enlarge the approved domain. Model, context and GPU-memory fit therefore remain separate admission questions rather than reasons to relax classification or destination restrictions (C12).

The comparative measurement obligation is to evaluate matched roots across eligible alternatives and report admissions and useful completions by stratum. An earlier inference from aggregate local usefulness to sensitive-task coverage cannot be retained: 800 successes among 1,000 admitted local

roots are compatible with all 200 sensitive roots failing and all 800 other roots succeeding. Imposed scenario fractions must consequently be distinguished from estimated probabilities and realized outcomes. Paired sample sizes and uncertainty procedures remain unspecified, and descendant calls or repeated evaluations must not be treated as independent root-task observations (C12).

The proposed authorization design binds each operation to an immutable canonical manifest containing the home issuer, principal, project, purpose, payload and attachments, classification restrictions, destination, requested model/version, tool permissions, retention or contract profile, and resource ceilings. Changing the payload or destination requires a new operation linked to the same root; an approved retry receives a new attempt identifier under an unchanged operation. Idempotency keys are scoped to principal and project, with conflicting manifests rejected rather than merged. Identifiers do not confer authority, and sensitive manifests, status information and deduplication records remain within approved access-controlled domains (C11).

Classification validation precedes dispatch and applies to derived material as well as the initial input. Restricted prompts, outputs, embeddings, logs and caches retain applicable restrictions unless a campus-authorized process changes them; ambiguous classification fails closed. Each tool disclosure is a separate operation, so an agent cannot approve its own destination. Protocol normalization must reject unsupported fields, unenumerated attachments or altered content rather than silently removing semantics that were material to authorization (C11).

The central refinement is a durable home-campus transaction authority that orders policy changes, identity and project revocations, cancellation, attempt transitions and relevant ledger updates. Preliminary approval and financial reservation are advisory with respect to transmission: only the final CommitDispatch(A) transaction creates a dispatch grant. Its commit point, $L(A)$, rechecks current authority, classification, exact manifest binding, destination and retention approval, model and tool permissions, reservation coverage and the designated controlled egress path. The same transaction records the protected audit event and consumes the attempt's one-use issuance right (C11).

Revocation is defined relative to that commit point, not eventual delivery. An applicable revocation R_v ordered before $L(A)$ must prevent the grant; if $L(A)$ precedes R_v, the grant authorizes only that immutable attempt, including possible later transmission by its already-designated sender. A revocation-effective acknowledgement is returned only after the revocation commits, so an invocation beginning after that acknowledgement cannot be ordered before it; a partitioned replica cannot create new grants from cached approval. This is explicitly decision-time authorization, and a campus requiring no transmission, execution or retention after revocation must reject these semantics or require additional recipient-side enforcement (C11).

Financial admission is coupled to the same authority without being confused with task success. For each applicable user, project, department, campus or shared-payer scope k, the design maintains a cap B_k, settled charges S_k and outstanding reservations H_k; reserving attempt A atomically requires $S_k + H_k + q_{A,k} \leq B_k$ in every scope, where $q_{A,k}$ is an enforceable maximum liability. A failed check changes none of the scopes. The liability bound must cover all attributable charges, and every spending path—including tools, retries and separately approved API accounts—must be mediated; an unknown bound blocks paid dispatch, while scheduling priority cannot override admission checks (C11).

Uncertainty after commitment preserves the reservation rather than restoring spending headroom. A crash, ambiguous charged timeout, lost response or provider rejection without final billing evidence does not justify retransmitting the same attempt or releasing its unresolved liability. Cancellation closes the operation to future grants and requests best-effort termination, but does not establish remote cancellation or a refund; a paid retry requires explicit approval, fresh checks and a new reservation, and is blocked while its predecessor's liability remains unresolved.

Before commitment, release is permitted only through a transition that permanently prevents the attempt from committing; afterward, release depends on authoritative reconciliation or durably fenced proof of non-issuance (C11).

The analytical method is transition induction over this abstract machine. Starting within every cap, atomic reservation preserves the accounting inequality, dispatch commitment leaves reserved liability unchanged, and bounded settlement replaces a reservation with a charge no greater than its ceiling. Authorization preservation likewise depends on final commitment being the sole grant-creation transition and on the controlled adapter enforcing one-use issuance without uncertain retransmission.

These are conditional arguments: complete mediation, trustworthy classification bindings, linearizability, crash-safe issuance, sender fencing and reconciliation are implementation premises, not properties established merely by declaring database states (C11).

The proposed fault-injection component uses adversarial event schedules and independent packet observation to test the gap between transactional state and actual egress. Cases include stale credentials, ambiguous classification, malicious-tenant bypass, concurrent revocation or cancellation, WAN partition, replay, simultaneous reservations, crashes, provider rejection, ambiguous timeouts, duplicate settlement and contradictory billing evidence. Discriminating failures are a grant ordered after an applicable revocation, duplicate issuance, unauthorized boundary crossing or liability exceeding a cap.

The recorded schedules provide expected outcomes rather than executed results; they also expose the intended availability cost of retaining unresolved reservations, which can persist indefinitely (C11).

The broader empirical proposal is a reviewed 90-day pilot measuring matched root-level quality, provenance, workflow amplification, memory and context fit, queue and cold-start tails, deadline attainment, unserved demand, availability, fairness and administrative effort. Its sample sizes, uncertainty procedures and stop/go or inconclusive thresholds have not been approved, and no pilot outcomes are recorded. Evidence for an improvement would therefore require both independently inspected enforcement and comparative root-level measurements, rather than a higher assumed usefulness fraction or a larger pool of nominally available GPUs (C11; C12).

Together, these methods change the federation argument from a capacity-sharing proposition into a constrained comparison of completed research goals. Federation must demonstrate additional policy-permitted, rubric-passing, deadline-meeting roots relative to campus service with separately approved external access, while preserving the chosen authorization ordering and all applicable liability caps. More calls, cached approvals or nominal partner eligibility cannot substitute for that evidence (C11; C12).

Table 2: Proposed monthly workload and acceptance instruments under C12. Counts, deadlines and rubrics are study-design assumptions requiring campus approval, not observed workload or benchmark results. Every accepted root must also satisfy authorization and contain no critical factual or security error.

Accounting bin	**Monthly root tasks**	**Proposed deadline**	**Task-specific acceptance instrument**
Classroom coding	600	30 s	Held-out coding tests
Routine analysis	400	120 s	Known-result and unit checks
Demanding scientific reasoning	300	600 s	Blinded scientific review
Multimodal/long-context work	200	180 s	Labeled multimodal fixtures
Agentic multi-call workflows	200	900 s end-to-end	Root-goal and tool-boundary checks
Campus-only sensitive analysis	200	120 s	Frozen task rubric; no separate sensitive-analysis oracle specified
Reproducible batches	100	24 h	Complete provenance and specified numerical tolerance; no bitwise API replay assumption

Evidence: [Research session 1, artifact 5]; claims: C12

Claims discussed: C11, C12

6 Evidence limitations and uncertainties

The evidence limits the central argument at the boundary between conditional design claims and demonstrated campus service. The supplied material supports examining when authorization, liability accounting and useful-completion comparisons would be valid; it does not establish that those conditions hold in a deployment. Missing source evidence, unimplemented enforcement and absent workload measurements are therefore distinct limitations, rather than interchangeable reasons for uncertainty. None alone demonstrates that federation is infeasible, but together they preclude a finding that it provides useful, reliable access within campus policy and budget constraints. [C13–C17]

The proposed evaluation defines the workload as registered parent tasks spanning coding, routine analysis, scientific reasoning, multimodal or long-context work, agentic workflows and reproducible batches, with sensitive tasks retaining their additional domain restrictions. A parent task—not an individual model call, retry or reproducibility repeat—is the unit of analysis; paired comparison additionally requires permission to evaluate that task in both the gateway-plus-approved-external-access and federation arms. The proposed outcome oracle requires the final artifact to satisfy a frozen quality rubric, authorized processing, required provenance and an end-to-end deadline.

Two blinded assessors would score artifacts independently, with a third adjudicating disagreement, within a proposed 90-day study horizon whose enrollment must leave time for delivery and grading. These definitions delimit the intended reliability question, but the task-specific pass rules, approved thresholds and executable study protocol remain incomplete. [C16]

Source uncertainty prevents treating the motivating platform description as an independently established capability inventory. The supplied evidence does not contain inspected MindRouter documentation or the PEARC'26 paper, verified access dates and versions, dated API prices, enforceable charging terms, retention commitments or institutional approvals. Consequently, neither the feature descriptions nor any claim that they remain current can be certified from this material. Federation and frontier routing remain hypothetical extensions, not demonstrated MindRouter capabilities; the absence of source verification is a gap in the comparison, not evidence that the published descriptions are false. [C13]

The strongest interpretation of the enforcement argument is conditional: atomic transactions, complete financial mediation, trusted request binding and controlled one-use issuance could support the stated accounting and authorization properties if their premises were implemented. The corresponding objection is operational rather than merely documentary: concurrent authorization, revocation, dispatch, recovery and nested reservations have not been implemented, independently inspected or approved. A database state transition does not itself establish that a sender cannot replay a request, that every spending credential is mediated, or that an external endpoint observes the required retention and charging limits.

Moreover, unavailable earlier state and transition definitions prevent checking the claimed exact correspondence between the proposed implementation model and its abstract specification. [C14,

C17]

Revocation illustrates why the contractual meaning of a guarantee matters as much as its mechanism. The proposed specification orders revocation against the final authorization decision: revocation ordered first blocks a new grant, whereas a previously committed grant can permit its designated sender to transmit later. That is not a guarantee of no transmission, execution or retention after revocation. Likewise, the partition argument concerns fresh disconnected authorization without authoritative updates or an earlier request-specific grant; it does not establish the same restriction for reachable home-campus decisions or for an explicitly weaker revocation contract. Campus acceptance of these distinctions remains absent. [C14]

The analytical counterexamples below identify failures of particular shortcuts, rather than observed incidents or refutations of every reservation-based design. Their importance is that replay control, financial exposure and authorization ordering cannot substitute for one another. A repaired specification must preserve each relevant property simultaneously. [C14]

Even a corrected reservation argument depends on an operational test for final liability. A current zero-charge report, a quiet local log or the close of an invoice period does not establish that a delayed sender or already-transmitted request can no longer incur charges. Conversely, retaining the full unresolved liability avoids restoring unsupported spending headroom but can impair availability and fairness if uncertainty persists indefinitely. Thus hard-budget preservation cannot be assessed independently of timeout reconciliation, provider contracts and the service consequences of conservative holds. [C14]

The service and economic evidence is weaker than the conditional accounting evidence. Actual costs, model fit, usable capacity, workflow call amplification, latency distributions, sensitive-task usefulness and unserved demand remain unmeasured in the supplied material. The requested four-baseline comparison is therefore incomplete: an illustrative aggregate budget or admission calculation cannot rank campus-local inference, approved API access, a campus gateway with separately approved external access, and federation on quality, coverage and cost.

Procurement conclusions also depend on distinguishing sunk, incremental and fully loaded costs, including replacement timing, power and cooling, storage and networking, support, procurement and staff effort. Without that ledger, an apparent saving has no established campus-specific accounting basis. [C15]

The recorded case against federation is correspondingly narrow. If federation adds no usable peer service, preserves identical non-federation entitlements and useful service opportunities, supplies no offsetting benefit and incurs positive overhead, then the gateway alternative is cheaper on the stated cost basis. That conditional comparison is not evidence that campuses typically have no useful capacity to share. Nominal GPU capacity is nevertheless insufficient evidence of benefit: the capacity must also be authorized, reachable, compatible with the requested model and feasible within the task deadline. Both federation superiority and a general claim of its economic disadvantage exceed the available observations. [C15]

Aggregate usefulness also leaves a consequential coverage ambiguity. A total expected-completion target does not establish that sensitive tasks receive useful service, because favorable outcomes in another stratum can conceal failure in the sensitive stratum. The recorded repair supplies

explicit stratum-level usefulness assumptions, but those assumptions remain hypothetical rather than measured. Expected counts likewise do not guarantee realized delivery, and constant assumed success fractions cannot establish performance as queues, capacity or paid-access limits become binding. Scheduling weights alone supply neither an unconditional latency guarantee nor evidence of starvation freedom. [C15, C16]

Statistical uncertainty is not resolved by accumulating more calls or more nominal task executions. The proposed paired inference assumes independently and identically sampled parent-task episodes, or a separately justified independent-block design; shared outages, queues, caches, repeated templates and a diminishing common budget can violate those premises. Reproducibility repeats assess repeatability but do not enlarge the independent primary sample. The supplied dependence counterexample makes the limitation explicit: if every paired outcome difference is the same shared random value, increasing the number of recorded tasks does not produce the variance reduction claimed under independence.

A controlled paired result would also apply only to its declared eligible population, not automatically to sensitive tasks prohibited from paired evaluation or to all live campus demand. [C16]

The decision criterion was substantively tightened rather than retrospectively validated. An exploratory rule requiring an estimated improvement of at least 10 percentage points and an interval excluding zero does not establish that the true improvement exceeds 10 percentage points; the later proposal instead requires a confidence bound above that margin. The supplied finite-sample planning argument is conditional on its specified sampling model and hypothetical alternative, not evidence of observed benefit, affordable enrollment or achievable completion within the study horizon.

Shared dependence, multiple confirmatory strata, safety stopping and resource constraints remain consequential qualifications, while absolute quality, security, availability and administrative gates require separate judgments. [C16]

External evidence capable of changing these conclusions must address the disputed premise rather than merely repeat the conditional argument. Source and contract inspection would establish what services and liabilities are actually available; implementation inspection and boundary fault injection, with independent invocation and ledger observations, could test enforcement; matched workload outcomes and a non-overlapping cost ledger could test comparative benefit. No such fault-injection results or campus benchmark are supplied, and zero observed violations in a finite test would provide bounded support rather than proof of general safety.

The evidence therefore changes the central argument from a claim of federation advantage to a conditional comparison: federation's additional useful service must be demonstrated alongside policy enforcement and net cost, while campus-only service with separately approved external access remains an unrejected alternative. [C13–C17]

Table 3: Analytical counterexamples and scope limits for proposed enforcement claims. These are hypothetical schedules and specification objections, not measured failures.

Claim being limited	**Supplied counterexample or boundary**	**Consequence for interpretation**
Timeout or cancellation makes a reservation safe to release.	With a cap of 10, releasing an unresolved liability of 6 and admitting another liability of 6 permits total exposure of 12.	Elapsed time or cancellation alone cannot establish financial finality; unresolved exposure must remain covered.
A fresh fenced namespace makes stale-backup recovery financially safe.	A restored backup omits an old committed liability of 6. Fresh identifiers prevent replay, but admitting a new liability of 6 still exceeds a cap of 10.	Replay isolation must be combined with preservation or conservative reconstruction of outstanding liabilities. The namespace-only recovery alternative is rejected.
Logical fencing prevents all later transmission.	A database fence can block new commits without revoking an already-authorized continuation, stopping an existing socket or recalling packets beyond campus control.	Decision-time authorization must not be presented as immediate physical egress revocation; stronger semantics require additional enforcement and campus acceptance.

Evidence: [Research session 1]; claims: C14, C17 *Claims discussed:* C13, C14, C15, C16, C17

7 Open questions and next steps

The next decision is whether federation can demonstrate an additional, policy-permitted research capability that justifies extending the trust boundary beyond campus service plus separately approved external access. The unresolved comparison is therefore not simply between more and fewer GPUs: it must establish whether approved remote capacity produces useful completions that the simpler arrangement cannot deliver, after accounting for service losses and administrative overhead. The evidence supports making federation conditional on that demonstration, rather than treating it as the default architecture. [C18, C22]

Source verification is the first prerequisite for specifying that comparison accurately. The MindRouter documentation and PEARC'26 short paper still require retrieval with actual access dates and versions, followed by verification of the attributed protocol translation, WDRR scheduling, quotas, audit logging, telemetry and campus-service scope. The project-reported review date of 2026-10-31 is historical information, not a newly established access date. Published descriptions, independently measured behavior and proposed extensions must remain separate; federation, frontier routing and the proposed financial controls cannot be credited to the existing gateway on the

supplied evidence. [C21]

Institutional decisions must then determine whether the proposed authorization semantics are acceptable, rather than leaving this question to implementation. Campus security and data owners need to approve classification inheritance for prompts, outputs, logs, embeddings and caches, together with identity, approved domains, retention and handling of in-flight work; legal and procurement review must establish recipient obligations, audit rights and cross-campus agreements. A particularly consequential choice is whether authorization at the authoritative decision point is sufficient, even though an earlier grant may transmit after a later revocation.

Campuses requiring stronger restrictions on subsequent transmission, execution or retention must require additional enforcement and recipient cooperation, or reject that specification. [C20]

Implementation work should address the gap between an authorization record and controlled execution. The transaction schema, immutable translated request, sender continuation, retry suppression, replay retention and complete mediation all require implementation and independent inspection. Recovery must preserve outstanding financial exposure as well as reject replay: fresh credentials or a new identifier namespace do not repair a stale backup that has forgotten liabilities. The recorded counterexample—an omitted liability of 6 followed by a new liability of 6 under a cap of 10—rules out that recovery shortcut without refuting the conditional reservation design. [C19]

The release of uncertain reservations requires an equally explicit operational rule. A timeout, cancellation acknowledgement or current report of zero charges cannot establish final liability if an already authorized sender could still act later. Before restoring spending headroom, the implementation must demonstrate terminalization of that possibility or obtain an authoritative, enforceable final-liability guarantee covering delayed requests and every applicable financial scope. Procurement review is therefore part of instantiating the hard-budget premise, not merely a later purchasing task; the same requirement applies to separately approved API accounts if their spending is included in the claimed cap. [C19, C20]

The empirical comparison should be organized around a reviewed ninety-day pilot of campus-local inference, approved API-only access, a campus gateway with separately approved external access, and the proposed federation extension. Its workload should explicitly cover coding, routine analysis, demanding scientific reasoning, multimodal or long-context work, agentic workflows, sensitive-data tasks and reproducible batches. The unit of analysis should be a registered parent research objective with fixed inputs, classification, requested artifact, deadline, provenance requirements and resource envelope; calls, tools, retries and reproducibility repeats remain descendants rather than additional successes.

Public or synthetic inputs should be used until the relevant campuses approve real-data processing, and enrollment must end early enough to complete the longest delivery deadline and adjudication within the pilot horizon. [C18, C20]

Before execution, faculty and students should approve task-specific pass rules and critical-failure criteria. The proposed outcome oracle is conjunctive: a completion must use authorized processing, satisfy the frozen quality rubric and provenance requirements, and deliver the final artifact within its end-to-end deadline. Registered coding tests, numerical checks, expert scientific assessment, task-specific multimodal references, agent objective and tool-boundary checks, and batch rerun

tolerances provide distinct evaluators for these requirements.

The proposed human evaluation uses two blinded assessors with a third adjudicator for disagreement; latency includes approval handling, queues, cold starts, tools, retries and recovery, rather than ending at the first streamed token. [C18]

The pilot also needs an approved population and uncertainty procedure before it can support a federation-benefit claim. The paired gateway–federation comparison should retain identical non-federation entitlements and a common registered denominator, treating queue rejection, model-fit failure, budget exhaustion and outages as outcomes rather than favorable exclusions. Repeated executions of the same parent task must not be counted as independent observations; independent-pair calculations require the declared sampling assumptions, while an independent-block analysis targets a block-average effect that need not equal the task-weighted effect.

The proposed stronger decision rule requires a confidence-interval lower endpoint exceeding 0.10 for the absolute useful-completion gain, not merely a point estimate of that size with an interval excluding zero. The earlier exploratory target of thirty tasks per stratum does not justify enrollment: exact power, precision, multiplicity and an affordable fixed sample remain to be established, with an explicit inconclusive outcome. [C18, C22]

Coverage and service measurements must remain disaggregated enough to expose who benefits and who remains unserved. In particular, sensitive-task usefulness requires its own evidence: aggregate local success cannot establish sensitive coverage, and the repaired expected-count scenario is not a realized-service guarantee. Measurements should distinguish policy eligibility, actual admission and useful completion by classification and task requirements, alongside heterogeneous GPU and context fit, compatible capacity, queue and cold-start tails, peak demand, maintenance, and batch versus interactive service.

Fairness assessment still requires an approved reference allocation and observation window; scheduling weights alone do not establish starvation freedom. [C18]

Fault injection should test the implementation's policy and financial boundaries separately from the workload benchmark. The proposed schedules include partitions, stale credentials or policy, malicious-tenant bypass, ambiguous classification, provider rejection, uncertain charged timeouts, cancellation, concurrent reservations and recovery. Independent egress and provider-side invocation observations should be compared with authorization and ledger state, rather than relying on response codes alone.

These studies should also quantify the availability and fairness costs of retained uncertainty holds; zero observed violations would provide bounded test evidence, not a universal safety result, and an early safety stop should leave the efficacy claim inconclusive. [C18, C19, C20]

The economic decision requires campus evidence on the same workload, horizon and accounting basis as the service comparison. Local ledgers should separate sunk, incremental and fully loaded costs while covering capital and replacement, utilization and idle capacity, power and cooling, storage, networking, support, procurement and operations staff; purchase cash flow and depreciation must not be counted twice. API ledgers need dated input, output and cached-token rates, complete workflow amplification, retries, tools, embeddings, egress, commitments and contractual charging bounds, while federation adds its actual integration, agreement and administrative costs.

Sensitivity analysis should vary demand, utilization, task mix, capacity and prices within feasible service ranges, rather than extrapolating constant useful-completion fractions through saturation or treating allocation ceilings as spending. [C18, C20, C22]

The deployment recommendation should follow these independent safety, coverage, service and administrative gates, not substitute an aggregate economic score for them. Computing staff must supply capacity and operational-effort evidence, security and data owners must define enforceable boundaries, procurement must resolve charging and retention terms, and faculty and students must approve usefulness, reproducibility, deadlines and fairness requirements. If federation supplies no additional usable, permitted service and no offsetting benefit while adding positive overhead on the chosen cost basis, the simpler gateway arrangement is cheaper under those stated conditions; whether actual campuses satisfy them remains a measurement question.

The evidence thus changes the central argument from a presumption that pooled capacity improves access to a requirement that federation earn its additional trust boundary through demonstrated useful service within approved policy and budget constraints. [C18, C20, C22]

Claims discussed: C18, C19, C20, C21, C22

8 Algorithms, results, and limitations

8.1 C1 – Unresolved

Under the partition model stated below, a disconnected federation admission point cannot both authorize new requests using cached campus approval and guarantee immediate enforcement of home-campus revocation. A safe design must deny or defer those requests, regain authoritative authorization, or explicitly adopt weaker, campus-approved lease semantics. This is a logical limitation of the proposed extension, not evidence of a defect in MindRouter. Recorded at claim_or_question. The saved research contains this proposition, but its proof, empirical status, and interpretation require independent verification. *Evidence:* [Research session 1, artifact 1]

8.2 C2 – Unresolved

Under the accounting and enforcement assumptions listed below, a campus-controlled, atomic reservation ledger preserves user, project, department and campus spending caps even when calls time out or are cancelled. Unresolved calls must retain their reservations; releasing a reservation merely because a timeout or cancellation occurred is unsafe. This safety property does not establish task quality, availability or starvation freedom, and does not require federation. Recorded at claim_or_question. The saved research contains this proposition, but its proof, empirical status, and interpretation require independent verification. *Evidence:* [Research session 1, artifact 3]

8.3 C3 – Unresolved

The supplied domains share authorized root-task accounting and enforceable nested liabilities. The source additionally assumes linearizable home authority, trusted manifest binding, mediated one-use issuance and campus acceptance of decision-time authorization. The historical target original is unavailable. Recorded at portfolio_decision/cross_strategy_reviews/0/transfer_review/assumption_comparison. The saved research contains this proposition, but its proof, empirical status, and interpretation require independent verification. *Evidence:* [Research session 1]

8.4 C4 – Unresolved

The source's sampling and paired-pilot assumptions do not establish sender safety. Its partition rationale matches only fresh disconnected decisions without authoritative information or an earlier grant, not reachable home decisions or explicitly weaker revocation contracts. Recorded at portfolio_decision/cross_strategy_reviews/5/transfer_review/assumption_comparison. The saved research contains this proposition, but its proof, empirical status, and interpretation require independent verification. *Evidence:* [Research session 1]

8.5 C5 – Unresolved

The source uses hypothetical workload and cost assumptions, including zero additional usable peer capacity. The target's recorded scope concerns fresh disconnected authorization without revocation information or prior request-specific approval. Only request-specific authorization eligibility overlaps; the target original is unavailable. Recorded at portfolio_decision/cross_strategy_reviews/2/transfer_review/assumption_comparison. The saved research contains this proposition, but its proof, empirical status, and interpretation require independent verification. *Evidence:* [Research session 1]

8.6 C6 – Unresolved

With n_G,local,S=200, the expected sensitive gate μ_G,S≥160 requires q_G,local,S≥0.80. If sensitive admissions instead equal a_S≤200, the condition is a_S q_G,local,S≥160; it is impossible when a_S<160. To retain the worksheet's 800 expected local completions, a simple explicit repair is q_G,local,S=q_G,local,N=0.80 with 200 and 800 admissions respectively. More generally, retaining the aggregate 800 and assuming q_G,local,S≥0.80 requires q_G,local,N=(800-200q_G,local,S)/800. The sensitive premise is additional and hypothetical, not implied by aggregate q_L. Recorded at investigator_output/reasoning_steps/3/argument. The saved research contains this proposition, but its proof, empirical status, and interpretation require independent verification. *Evidence:* [Research session 1, artifact 8]

8.7 C7 – Unresolved

The supplied project description attributes protocol translation across OpenAI/Ollama/Anthropic clients and Ollama/vLLM backends, WDRR scheduling, per-user quotas, audit logging and GPU telemetry to MindRouter's published campus gateway. These are starting descriptions, not independently measured capabilities in this session. Cross-campus federation, frontier routing, hierarchical financial reservations and the enforcement protocol below are proposed extensions, not asserted MindRouter features. The logical result concerns an abstract ledger, not a verified implementation. Recorded at investigator_output/reasoning_steps/0/argument. The saved research contains this proposition, but its proof, empirical status, and interpretation require independent verification. *Evidence:* [Research session 1, artifact 3]

8.8 C8 – Unresolved

The supplied project attributes protocol translation across OpenAI/Ollama/Anthropic clients and Ollama/vLLM backends, WDRR fair-share scheduling, per-user quotas, audit logging and GPU telemetry to MindRouter's published campus gateway. These are supplied descriptions of published claims, not independently measured facts here. The documentation URL is `https://mindrouter.ai/documentation.html` and the paper URL is `https://ai4ra.uidaho.edu/wp-content/uploads/2026/04/Sheneman_MindRouter_PEARC26_Short_Paper.pdf`. The supplied starting-reference review date is 2026-10-31; this session has no access date because neither source was retrieved. No changed claim is verified. Define L as campus-local inference only; A as approved frontier API only; G as campus gateway plus separately approved external access; F as a hypothetical cooperative federation, with external routing only if separately authorized. Neither federation nor frontier routing is treated as an existing MindRouter feature. Recorded at investigator_output/reasoning_steps/0/argument. The saved research contains this proposition, but its proof, empirical status, and interpretation require independent verification. *Evidence:* [Research session 1, artifact 2]

8.9 C9 – Unresolved

The supplied extract of [Research session 1, artifact 4] provides the cycle-2 states and expected counterexamples; its original is not supplied and parts of the extract are incomplete. I use that extract as an unverified specification to refine, not a certified premise. The original [Research session 1, artifact 1] is pointer-only and has not been inspected. Retain two explicit constraints from the supplied cycle-2 text: a pre-revocation grant can transmit later under decision-time semantics; releasing on timeout can expose two liabilities of 6 against cap 10. The present design neither restores the stronger delivery guarantee nor reuses timeout release. Recorded at investigator_output/reasoning_steps/0/argument. The saved research contains this proposition, but its proof, empirical status, and interpretation require independent verification. *Evidence:* [Research session 1, artifact 7]

8.10 C10 – Unresolved

The supplied bounded extract of [Research session 1, artifact 3] was reviewed, not its saved original. It contains a conditional reservation argument, counterexamples to releasing ambiguous liabilities and to unconditional WDRR latency guarantees, and an underspecified pilot. Preserve those negative constraints: do not release a hold merely after timeout/cancellation, and do not infer starvation freedom from scheduling weights. Replace its exploratory target of 30 tasks per stratum and its point-estimate-at-least-10-points/interval-excluding-zero gate with the stronger confidence-bound gate below. This is a deliberate tightening, not an interpretation that the earlier gate already established a true benefit exceeding 10 points. All recorded source, approval, implementation, cost and empirical obligations remain unresolved. Recorded at investigator_output/reasoning_steps/0/argument. The saved research contains this proposition, but its proof, empirical status, and interpretation require independent verification. *Evidence:* [Research session 1, artifact 6]

8.11 C11 – Unresolved

Use a proposed home-campus transaction authority that linearizes policy changes, identity/project revocations, operation cancellation, attempt transitions and all relevant ledger updates. Preliminary approval and reservation are not permission to transmit. The final CommitDispatch(A) transaction is the authorization ordering point L(A). An applicable revocation Rv ordered before L(A) must prevent that grant; L(A) ordered before Rv authorizes only that immutable attempt, even if its already-designated sender transmits later. Concurrent events take their committed order; a campus receives a revocation-effective acknowledgement only after its transaction commits. An invocation that begins after that acknowledgement cannot be ordered before revocation. A partitioned replica cannot commit new grants from cached approval. This chooses decision-time semantics explicitly; campuses requiring no transmission, execution or retention after revocation must reject this specification or require additional recipient/enforcement controls. Recorded at investigator_output/reasoning_steps/2/argument. The saved research contains this proposition, but its proof, empirical status, and interpretation require independent verification. *Evidence:* [Research session 1, artifact 4]

8.12 C12 – Unresolved

Use 100 distinct active researchers and 2,000 root tasks each month: coding 600, routine analysis 400, demanding scientific reasoning 300, multimodal/long-context 200, agentic workflows 200, campus-only sensitive analysis 200, reproducible batches 100. These are disjoint accounting bins, not a claim that sensitivity is a distinct technical task type. Proposed deadlines respectively are 30 s, 120 s, 600 s, 180 s, 900 s end-to-end, 120 s and 24 h. A useful completion must be authorized, meet its deadline, pass the frozen task rubric and contain no critical factual/security error. Rubrics use held-out coding tests, known-result/unit checks, blinded scientific review, labeled multimodal fixtures and agent root-goal/tool-boundary checks. Batch acceptance requires complete provenance and a specified numerical tolerance; it does not assume bitwise API replay. Calls, Investigator/Skeptic exchanges and retries remain descendants of one root. Correct refusal

is recorded separately, not counted as delivered research. These deadlines and rubrics require campus approval. Recorded at investigator_output/reasoning_steps/1/argument. The saved research contains this proposition, but its proof, empirical status, and interpretation require independent verification. *Evidence:* [Research session 1, artifact 5]

8.13 C13 – Limitation

Published capabilities, current rates, contractual terms and campus approvals lack inspected source evidence. Recorded at outcome/caveats/2. The saved research contains this proposition, but its proof, empirical status, and interpretation require independent verification. *Evidence:* [Research session 1]

8.14 C14 – Limitation

Concurrent authorization, revocation, dispatch, recovery and reservation enforcement are unimplemented and unapproved. Recorded at outcome/caveats/4. The saved research contains this proposition, but its proof, empirical status, and interpretation require independent verification. *Evidence:* [Research session 1]

8.15 C15 – Limitation

Actual costs, workload performance, capacity, sensitive-stratum usefulness and net federation benefit remain unknown. Recorded at outcome/caveats/3. The saved research contains this proposition, but its proof, empirical status, and interpretation require independent verification. *Evidence:* [Research session 1]

8.16 C16 – Limitation

The broader 90-day study protocol and decision thresholds remain incompletely operationalized. Recorded at outcome/caveats/5. The saved research contains this proposition, but its proof, empirical status, and interpretation require independent verification. *Evidence:* [Research session 1]

8.17 C17 – Limitation

Open obligations remain; inspect the saved artifacts before relying on this result. Recorded at outcome/caveats/6. The saved research contains this proposition, but its proof, empirical status, and interpretation require independent verification. *Evidence:* [Research session 1]

8.18 C18 – Unresolved

o96/o102/o125/o130 and earlier comparison duties remain open: complete the four-baseline workload, quality, provenance, latency, throughput, GPU-fit, fairness, administrative-burden and cost study, with dated prices, sunk/incremental/fully loaded accounting, budget sensitivities and a reviewed 90-day pilot. Recorded at investigator_output/open_obligations/4. The saved research contains this proposition, but its proof, empirical status, and interpretation require independent verification. *Evidence:* [Research session 1, artifact 7]

8.19 C19 – Unresolved

o94/o100/o122/o127 and related implementation duties remain open: implement and independently inspect the transaction schema, immutable translation, sender continuation, retry suppression, replay retention, restore fencing and complete mediation. The present sketch advances specification only. Recorded at investigator_output/open_obligations/0. The saved research contains this proposition, but its proof, empirical status, and interpretation require independent verification. *Evidence:* [Research session 1, artifact 7]

8.20 C20 – Unresolved

o93/o99/o123/o128 and related governance duties remain open: obtain campus acceptance of classification inheritance and revocation semantics, and security/legal/procurement validation of approved domains, retention, final billing evidence and enforceable liability ceilings. Recorded at investigator_output/open_obligations/2. The saved research contains this proposition, but its proof, empirical status, and interpretation require independent verification. *Evidence:* [Research session 1, artifact 7]

8.21 C21 – Unresolved

o92/o98/o124/o129 and earlier source duties remain open: retrieve both starting references, record actual access dates and versions, verify the reported protocol translation, WDRR, quotas, audit and telemetry claims, and distinguish published features from proposed federation/frontier routing. The project-reported 2026-10-31 review is not this session's access date. Recorded at investigator_output/open_obligations/3. The saved research contains this proposition, but its proof, empirical status, and interpretation require independent verification. *Evidence:* [Research session 1, artifact 7]

8.22 C22 – Unresolved

o6 and related federation-benefit duties remain open: establish a measured policy-permitted capability gain over campus service plus separately approved external access. Obtain the original target and peer artifacts before stronger evidence transfer; the supplied handoff extracts do not establish economic or statistical conclusions. Recorded at investigator_output/open_obligations/5. The

saved research contains this proposition, but its proof, empirical status, and interpretation require independent verification. *Evidence:* [Research session 1, artifact 7]

9 Limitations

- The saved record may contain unverified interpretations and unresolved evidence conflicts; these require independent assessment.

10 Open obligations

- Retrieve both starting sources, record actual access dates and versions, and verify the user-supplied characterization of protocol translation, WDRR, quotas, audit logging, telemetry, and on-campus scope. No federation or frontier-routing feature has been verified.
- Retrieve the MindRouter documentation and PEARC'26 paper, record actual access dates and versions, and verify the supplied feature descriptions and any changes. Obtain dated API rates, contractual charging bounds, retention terms and institutional approval; none is established here.
- Inspect both starting references, record actual access dates and versions, and verify the precise published MindRouter capabilities and any changed claims. Do not infer proposed federation or financial controls from the supplied gateway description.
- Source grounding remains unresolved: obtain both starting references, record actual access dates and versions, and verify protocol translation, WDRR, quotas, audit logging, telemetry and campus-only scope. The project-reported 2026-10-31 review date is not this session's access date; no changed claim has been verified.
- o34/o41/o51/o55 remain unresolved: retrieve both starting sources and their original saved context where needed, record actual dates/versions, verify published features and changes, and obtain dated provider prices, contracts, retention and approvals. Peer extracts are truncated and their omitted details remain unknown.
- Verify the MindRouter documentation and PEARC'26 paper with actual access dates and versions; inspect saved originals where needed. Source obligations o7/o56/o63/o68 remain open.
- o94/o100/o122/o127 and related implementation duties remain open: implement and independently inspect the transaction schema, immutable translation, sender continuation, retry suppression, replay retention, restore fencing and complete mediation. The present sketch advances specification only.
- Retrieve the two starting references and pointer-only originals before reusing their detailed claims or numerical ledger; record actual access dates/versions and obtain dated API prices, charging bounds, retention terms and institutional approvals. Source-related ledger obligations remain unresolved.

- Ask campus security and data owners to define classification inheritance, approved domains, revocation timing, in-flight handling, and acceptable lease semantics; ask procurement/legal to validate partner/provider retention, subprocessors, audit rights, and enforceable charge bounds.
- Collect a campus cost ledger that separates sunk, incremental and fully loaded cost, including replacement timing, power/cooling, storage/network, support, procurement and staff effort. Replace illustrative aggregate totals before making a procurement recommendation.
- Obtain campus security/legal decisions on data classification, derived artifacts, identity federation, retention, incident response, contracts and permitted domains. Determine whether remote execution can actually enforce them.
- Campus/security/legal/procurement decisions remain required for classification inheritance, decision-time versus stronger revocation semantics, in-flight handling, destination and retention approval, final billing evidence and enforceable liability bounds.
- o35/o42 remain unresolved: replace every hypothetical ledger row with campus evidence, validate avoidability and staff scopes, inventory heterogeneous GPUs, and reconcile purchase/replacement cash flow with depreciation and stranded capacity.
- Obtain campus security/legal/procurement approvals, enforceable billing bounds and complete financial/egress mediation; test policy freshness, retention, cancellation and recovery. Obligations o8–o10/o57–o58/o65–o69 remain open.
- o95/o101 and related testing duties remain open: execute the fault schedules and independent egress/provider-observation tests, including adversarial tenants and stale credentials; quantify availability and fairness losses from uncertainty holds.
- Replace hypothetical costs with campus evidence, preserving sunk, incremental, fully loaded and acquisition/replacement cash-flow distinctions. Validate federation overhead and any offsetting benefit rather than assuming them.

11 Frozen obligation reconciliation

Unlisted obligations are not presumed resolved. The archived research snapshot retains complete source references and the full reconciliation.

- Retrieve the MindRouter documentation and PEARC'26 paper, record actual access dates and versions, and verify the supplied feature descriptions and any changes. Obtain dated API rates, contractual charging bounds, retention terms and institutional approval; none is established here. *(Listed as open in paper; 7 saved source reference(s))*
- Collect a campus cost ledger that separates sunk, incremental and fully loaded cost, including replacement timing, power/cooling, storage/network, support, procurement and staff effort. Replace illustrative aggregate totals before making a procurement recommendation. *(Listed as open in paper; 7 saved source reference(s))*

- Measure matched task quality, complete workflow token/call amplification, latency distributions, fit/cold-start behavior, capacity, unserved demand and provenance limits. Estimate uncertainty and sensitivity to task mix, utilization, API prices and demand rather than extrapolating constant success fractions through saturation. *(Not listed by author; not presumed resolved; 7 saved source reference(s))*

- Establish whether all paid calls, tools and retries can be assigned enforceable maximum charges and whether timeout reconciliation and nested quotas preserve the reservation invariant under concurrency and failures. This requires implementation testing and contract review. *(Not listed by author; not presumed resolved; 7 saved source reference(s))*

- Resolve campus identity, approved-domain classification, derived-artifact handling, retention, audit, incident and cross-campus agreement requirements before any sensitive or federated testing. *(Not listed by author; not presumed resolved; 7 saved source reference(s))*

- Run the bounded pilot and test whether federation adds enough useful completions to justify its net overhead. Safety fault injection with zero observed violations is only bounded evidence, not proof of general safety. *(Not listed by author; not presumed resolved; 7 saved source reference(s))*

- Ask campus computing about measured capacity and on-call effort; security about classification and derived-data boundaries; procurement about price changes, commitments and ambiguous charges; faculty about quality and reproducibility; students about access, deadlines and fairness. Their answers may reverse the preferred baseline. *(Not listed by author; not presumed resolved; 7 saved source reference(s))*

- o15: Retrieve and date the starting sources; obtain current rate, charging, retention and approval evidence. *(Not listed by author; not presumed resolved; 8 saved source reference(s))*

- o16: Replace aggregate illustrative costs with a non-overlapping campus ledger separating sunk, incremental and fully loaded costs. *(Not listed by author; not presumed resolved; 8 saved source reference(s))*

- o17: Establish matched quality, workflow amplification, latency, capacity, provenance and sensitivity evidence. *(Not listed by author; not presumed resolved; 8 saved source reference(s))*

- o18: Establish enforceable charge bounds and test concurrent reservations, reconciliation and nested quotas. *(Not listed by author; not presumed resolved; 8 saved source reference(s))*

- o19: Resolve classification, derived-artifact boundaries, identity, retention, audit and cross-campus authority. *(Not listed by author; not presumed resolved; 8 saved source reference(s))*

- o20: Execute the approved bounded pilot and assess federation's net benefit with uncertainty and independent safety gates. *(Not listed by author; not presumed resolved; 8 saved source reference(s))*

- o21: Obtain computing, security, procurement, faculty and student input. *(Not listed by author; not presumed resolved; 8 saved source reference(s))*

- Material, recoverable scope gap: an accounting threshold and aggregate synthetic example do not yet provide the requested four-baseline cost, quality, coverage and capacity comparison. *(Not listed by author; not presumed resolved; 6 saved source reference(s))*
- Material, recoverable implementation gap: the budget invariant is conditional on enforceable liability bounds and atomic nested-ledger transitions. It is not evidence that provider billing, cancellations, timeouts or concurrent quotas satisfy those premises. *(Not listed by author; not presumed resolved; 6 saved source reference(s))*
- Material, recoverable safety gap: the architecture does not yet specify policy freshness/revocation and reservation ownership across admission, dispatch, retries and reconciliation. No deployment-safety conclusion follows. *(Not listed by author; not presumed resolved; 6 saved source reference(s))*
- Evidence limitation, not a contradiction: source descriptions, current rates, retention terms, campus policy and workload performance remain unverified. *(Not listed by author; not presumed resolved; 6 saved source reference(s))*

213 further distinct obligations are retained in the archived reconciliation.

12 Conclusions

The investigation of Sovereign AI Federation v2 yields a provisional, source-linked account rather than an independently verified experimental or formal result.

One central recorded finding is: With n_G,local,S=200, the expected sensitive gate μ_G,S≥160 requires q_G,local,S≥0.80. If sensitive admissions instead equal a_S≤200, the condition is a_S q_G,local,S≥160; it is impossible when a_S<160. To retain the worksheet's 800 expected local completions, a simple explicit repair is q_G,local,S=q_G,local,N=0.80 with 200 and 800 admissions respectively. More generally, retaining the aggregate 800 and assuming q_G,local,S≥0.80 requires q_G,local,N=(800-200q_G,local,S)/800. The sensitive premise is additional and hypothetical, not implied by aggregate q_L. Its validity remains conditional on the saved premises and outstanding review.

Other recorded findings could qualify or conflict with this interpretation: The supplied project description attributes protocol translation across OpenAI/Ollama/Anthropic clients and Ollama/vLLM backends, WDRR scheduling, per-user quotas, audit logging and GPU telemetry to MindRouter's published campus gateway. These are starting descriptions, not independently measured capabilities in this session. Cross-campus federation, frontier routing, hierarchical financial reservations and the enforcement protocol below are proposed extensions, not asserted MindRouter features. The logical result concerns an abstract ledger, not a verified implementation.

The supplied project attributes protocol translation across OpenAI/Ollama/Anthropic clients and Ollama/vLLM backends, WDRR fair-share scheduling, per-user quotas, audit logging and GPU telemetry to MindRouter's published campus gateway. These are supplied descriptions

of published claims, not independently measured facts here. The documentation URL is `https://mindrouter.ai/documentation.html` and the paper URL is `https://ai4ra.uidaho.edu/wp-content/uploads/2026/04/Sheneman_MindRouter_PEARC26_Short_Paper.pdf`. The supplied starting-reference review date is 2026-10-31; this session has no access date because neither source was retrieved. No changed claim is verified.

Define L as campus-local inference only; A as approved frontier API only; G as campus gateway plus separately approved external access; F as a hypothetical cooperative federation, with external routing only if separately authorized. Neither federation nor frontier routing is treated as an existing MindRouter feature. The supplied extract of [Research session 1, artifact 4] provides the cycle-2 states and expected counterexamples; its original is not supplied and parts of the extract are incomplete. I use that extract as an unverified specification to refine, not a certified premise. The original [Research session 1, artifact 1] is pointer-only and has not been inspected.

Retain two explicit constraints from the supplied cycle-2 text: a pre-revocation grant can transmit later under decision-time semantics; releasing on timeout can expose two liabilities of 6 against cap 10. The present design neither restores the stronger delivery guarantee nor reuses timeout release. Their compatibility is not independently established.

The principal remaining evidence boundary is: Published capabilities, current rates, contractual terms and campus approvals lack inspected source evidence.

The next unresolved question is: Retrieve both starting sources, record actual access dates and versions, and verify the user-supplied characterization of protocol translation, WDRR, quotas, audit logging, telemetry, and on-campus scope. No federation or frontier-routing feature has been verified.

A stronger conclusion requires resolving the identified objections, checking the exact archived evidence and conducting any proposed verification before making empirical or deployment claims.

Claims discussed: C6, C13, C18, C7, C8, C9

Evidence and authorship note

Snapshot SHA-256: 552f9458e600d3e7d0b3aa6c1dd105cc62dab4967db652123d2f134783b76f34 Generated by an AI author from the cited saved research snapshot. This document is not independent verification. No external bibliography was added.

Validation warnings

- Evidence indexing omitted 9706 material saved research field fragment(s). Those fragments were not reviewed for this paper; the interpretation is provisional where the omitted evidence may matter. Consult the frozen source snapshot and omission ledger for full details.
- The AI planning stage did not yield a usable outline. A conservative, source-linked outline was reconstructed from saved evidence without repeating the planning request; review the

resulting paper's structure.

- The fallback outline selected representative indexed sources; the complete frozen research snapshot remains available for review.
- The evidence index explicitly omitted 9706 saved field fragment(s); see the generation coverage ledger and review the complete archived snapshot.
- Structured paper indexing selected 390 source-linked text fragments from 13 archived sources without AI extraction calls. The complete frozen snapshot remains archived; this bounded index does not constitute review of every saved field.
- Citation-like external references appear in the draft, but this paper has no verified external-source catalog; treat them as unsupported.
- The frozen evidence contains open proof/research obligation(s) requiring reconciliation: 215 item(s) are not listed by the author and are not presumed resolved; 0 author-listed item(s) have no exact frozen-record match. See the obligation reconciliation.
- The methods narrative has no linked claim IDs; check it for unsupported factual statements.
- AI-authored analysis is not independent formal verification; model agreement does not establish proof.

Typesetting verification note

Needs review: TeX verification is incomplete. The authored research content and safe accepted typesetting corrections were retained. Review the saved proofreading diagnostics before relying on the layout.